# Circuit Theorems Assignment

ECE2107 Circuits & Systems · assignment marathon · batch 1: Part A Superposition (5.1–5.14)

## How to use this

Every question follows the same order:

1. **Question**, with the figure taken straight from the book.
2. **Try it first** (yellow). Cover the rest of the question and solve it.
3. **Hint: what's new** (orange). Only when the question brings an idea you haven't met earlier in this PDF. If you're stuck, read it and try again.
4. **Solution** in compact assignment form: Given → To find → each step with its intermediate values → boxed answer. Copy it into your submission, or compare it line by line with your attempt to find the step that went wrong.
5. **Side note** (grey) or **book error** (red), only when there is something worth knowing.

The tag next to each question number shows how much teaching it carries: `teach` full explanation · `new idea` one new thing · `practice` plain solution.

If you couldn't solve one, just tell me its number. Nothing else needs marking.

## What's in it (full version)

| Part | Book | Questions | Count |
|---|---|---|---|
| A | Ghosh & A K Chakraborty, *Circuit Theory and Networks*, ch. 5 (**solved**) | Superposition 5.1–5.14 · Thevenin/Norton 5.15–5.34 · Max power 5.35–5.44 | 44 |
| B | Alexander & Sadiku, *Fundamentals of Electric Circuits* 5e (**unsolved**) | Practice problems + end-of-chapter: ch. 4 (superposition, source transformation, Thevenin/Norton, max power), ch. 10 (the same in AC), ch. 11.3 (AC max power) | 107 + practice problems |

Scope follows the official MUJ ECE2107 syllabus: superposition, Thevenin's and Norton's theorems, maximum power transfer, DC and AC, plus source transformation.

**Checking.** Every answer was worked out again and confirmed by a full circuit solve with all sources on (`src/verify.py`), including each case's partial answer. Matches with the book are marked "= book". Where the book is wrong, the correct answer is given and a red box explains the book's slip.

# Part A · Chakraborty (solved problems)

Ghosh & A K Chakraborty, *Circuit Theory and Networks* 4e, ch. 5 "Network Theorems", Solved Problems

## A · Superposition theorem

KVL, KCL, Ohm's law (you have these) → Nodal / mesh: all sources at once → **Superposition: one source at a time** → Thevenin / Norton (next)

**Where this sits.** You can already solve any circuit with nodal or mesh analysis, taking every source at once. Superposition breaks the same job into smaller circuits, **one source at a time**, which is often quicker by hand. It is also the reason Thevenin's theorem works, which comes next.

**Three terms first.**

- **Linear circuit:** made of R, L, C, linear dependent sources and independent sources. Doubling every source doubles every current and voltage.
- **Independent source:** a fixed voltage or current (a battery, a 2 A source, 10 cos 4t).
- **Dependent source** (diamond symbol): its value follows some other voltage or current in the circuit, for example $2V_x$.

**Statement.** In a linear circuit with several independent sources, any current or voltage is the **algebraic sum** of the values caused by **each independent source acting alone**.

**Switching a source off** means setting its value to zero:

| Source | Switched off it becomes | Why |
|---|---|---|
| Ideal voltage source | **short circuit** | 0 V across it |
| Ideal current source | **open circuit** | 0 A through it |
| Dependent source | **never switched off** | it is not a source on its own; it follows the circuit |

Superposition adds currents and voltages only, **not power** (P goes as $I^2$, and $(I' + I'')^2 \neq I'^2 + I''^2$).

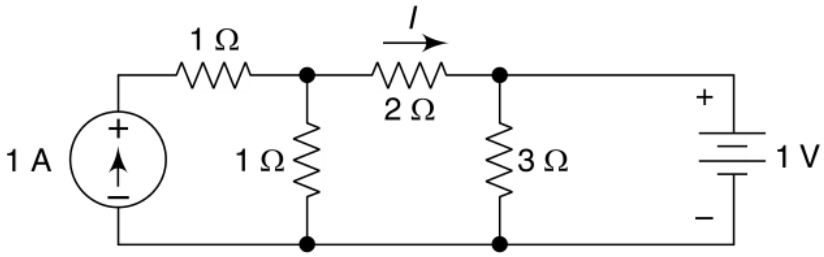
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### A1 · Problem 5.1

Superposition · DC **teach**

**QUESTION**

Find the current I in the circuit shown, using the superposition theorem.



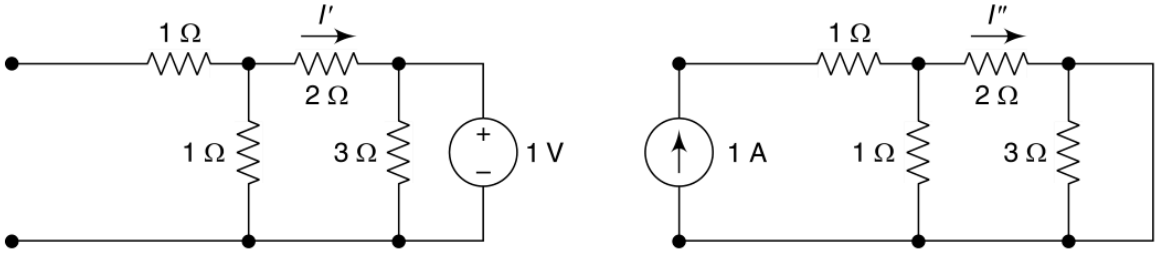

I flows left to right through the 2 Ω resistor.

**TRY IT FIRST**

Cover everything below. Draw two circuits, 1 V alone and 1 A alone, find the current in the 2 Ω in each, and decide each one's sign before adding.

**HINT: WHAT'S NEW**

- **The whole method:** two sources, so solve **two small circuits** and add.
- **Switching off:** the 1 V source becomes a short; the 1 A source becomes an open.
- **Signs:** measure each part against the arrow drawn for I. A part flowing the other way is negative.




(i) 1 V acting alone (1 A opened) · (ii) 1 A acting alone (1 V shorted)

**Given:** 1 A source, 1 V source; 1 Ω (series), 1 Ω (shunt), 2 Ω, 3 Ω.

**To find:** I through the 2 Ω (left → right).

**Case (i): 1 V alone** (1 A opened). The series 1 Ω carries nothing. The 3 Ω sits directly across the 1 V, so it changes nothing else. The 1 V drives current through 2 Ω + 1 Ω (shunt):

$$\text{current} = \frac{1}{2+1} = \frac{1}{3}\ \text{A (right → left)} \qquad \Rightarrow \qquad I' = -\frac{1}{3}\ \text{A}$$

**Case (ii): 1 A alone** (1 V shorted, so the 3 Ω is shorted out). The 1 A divides between the 1 Ω shunt and the 2 Ω:

$$I'' = 1 \times \frac{1}{1+2} = \frac{1}{3}\ \text{A (left → right)}$$

**Superposition:** $I = I' + I'' = -\frac{1}{3} + \frac{1}{3} = 0$

**ANSWER**

**I = 0 A** (= book)

**WHY ZERO, AND TWO SHORTCUTS**

- Node check: $1 = \frac{V_A}{1} + \frac{V_A - 1}{2}$ gives $V_A = 1$ V, the same as the right-hand node, so no current flows in the 2 Ω.
- An element **in series with an ideal current source** never changes any other current (the left 1 Ω here).
- An element **in parallel with an ideal voltage source** never changes any other current (the 3 Ω here).

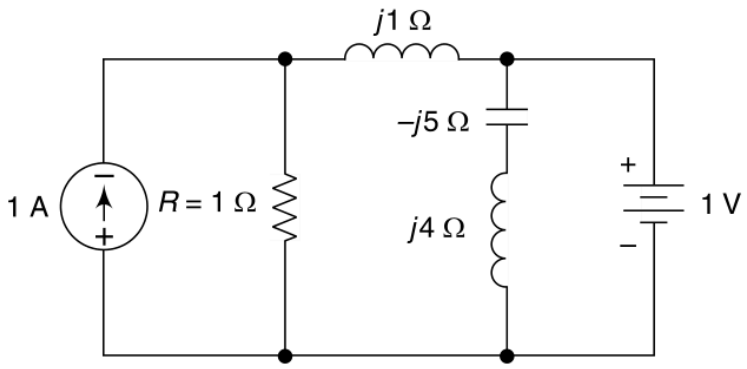
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## A2 · Problem 5.2

Superposition · AC **teach**

**QUESTION**

Calculate the voltage V across the resistor R by using the superposition theorem.



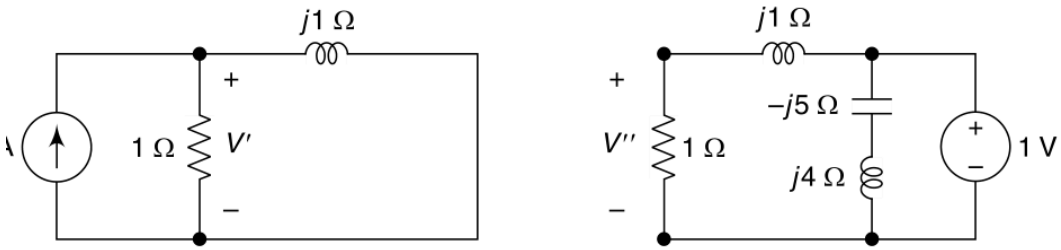

The book draws +/− signs inside the current source. Read it by its arrow: 1 A pushed up into the top node.

**TRY IT FIRST**

Which elements drop out in each case? Aim for V′ and V″ in the form $\frac{\text{something}}{1 + j}$.

**HINT: WHAT'S NEW**

- **AC (phasor) circuit.** j1 Ω is an inductor, −j5 Ω a capacitor. Superposition is unchanged; the arithmetic uses complex numbers.
- **Complex dividers:** the same formulas as for resistors, with Z in place of R.
- The −j5 + j4 branch is **across the 1 V source** (5.1's shortcut), so it never matters.




(i) 1 A acting alone · (ii) 1 V acting alone

**Given:** 1 A, 1 V; R = 1 Ω, j1 Ω, −j5 Ω, j4 Ω.

**To find:** V across R (+ at top).

**Case (i): 1 A alone** (1 V shorted; this also shorts out the −j5 + j4 branch). R ∥ j1 across the 1 A:

$$V' = 1 \times \frac{1 \times j1}{1 + j1} = \frac{j}{1 + j}\ \text{V}$$

**Case (ii): 1 V alone** (1 A opened). j1 and R in series across 1 V:

$$I'' = \frac{1}{1 + j} \qquad \Rightarrow \qquad V'' = 1 \times I'' = \frac{1}{1 + j}\ \text{V}$$

**Superposition:** $V = \frac{j}{1 + j} + \frac{1}{1 + j} = 1\ \text{V}$

**ANSWER**

**V = 1∠0° V** (= book)

**CHECK**

Node equation at the top node: $1 = \frac{V}{1} + \frac{V - 1}{j1}$ gives V(1 + j) = 1 + j, so V = 1 V ✓.

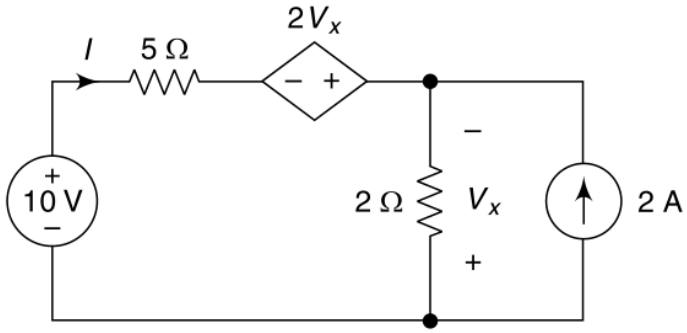
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## A3 · Problem 5.3

Superposition · dependent source **teach**

**QUESTION**

Use the superposition theorem on the circuit shown to find I.



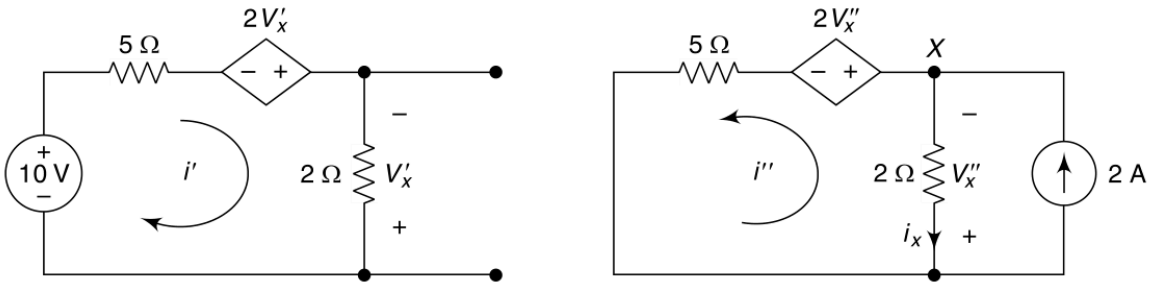

$2V_x$ is a voltage-controlled voltage source. $V_x$ is across the 2 Ω with + at the bottom.

**TRY IT FIRST**

Write $V_x'$ in terms of the loop current in case (i). For case (ii), write one KCL at the top of the 2 Ω and one KVL around the left loop. Expect elevenths.

**HINT: WHAT'S NEW**

- **Dependent source:** it **stays in every case**. Only the 10 V and the 2 A take turns.
- **Its value changes per case:** $2V_x'$ in case (i), $2V_x''$ in case (ii). Re-find $V_x$ in each sub-circuit.
- **Polarity of $V_x$:** + is at the **bottom** of the 2 Ω. Most wrong answers to this question come from this sign.




(i) 10 V alone (2 A opened) · (ii) 2 A alone (10 V shorted); the dependent source stays in both

**Given:** 10 V, 2 A, 5 Ω, 2 Ω, VCVS $2V_x$.

**To find:** I (out of the 10 V source, through the 5 Ω).

**Case (i): 10 V alone.** One loop, current i′ clockwise (= direction of I), flowing down the 2 Ω, so $V_x' = -2i'$.

$$10 + 2V_x' = 5i' + 2i' \quad \Rightarrow \quad 10 - 4i' = 7i' \quad \Rightarrow \quad I' = i' = \frac{10}{11}\ \text{A}$$

**Case (ii): 2 A alone.** At node X: 2 A splits into $i_x$ (down the 2 Ω, so $V_x'' = -2i_x$) and i″ (leftward through the dependent source and 5 Ω).

$$\text{KCL at X:} \qquad 2 = -\frac{V_x''}{2} + i'' \qquad \ldots(1)$$

$$\text{Left loop:} \qquad 5i'' + 3V_x'' = 0 \qquad \Rightarrow \qquad i'' = -\frac{3}{5}V_x'' \qquad \ldots(2)$$

(X sits at $-V_x''$; the dependent source drops another $2V_x''$, so the 5 Ω has $-3V_x''$ across it.)

$$\text{(2) in (1):} \qquad 2 = -\frac{11}{10}V_x'' \qquad \Rightarrow \qquad V_x'' = -\frac{20}{11}\text{ V}, \qquad i'' = \frac{12}{11}\text{ A}$$

$i''$ runs anticlockwise, against I, so **$I'' = -12/11$ A**.

**Superposition:** $I = \frac{10}{11} - \frac{12}{11} = -\frac{2}{11}$ A

**ANSWER**

**$I = -2/11$ A ≈ −0.182 A** (= book). The minus sign means the current actually flows into the + of the 10 V source.

**NEVER SWITCH OFF A DEPENDENT SOURCE**

Shorting $2V_x$ as if it were independent gives a wrong answer. It stays in; only its value changes from case to case.

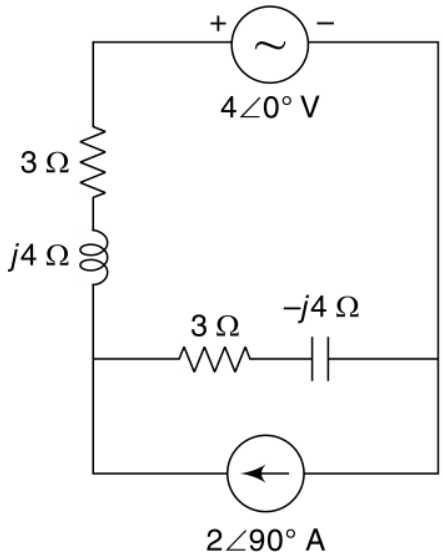
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## A4 · Problem 5.4

Superposition · AC **new idea**

**QUESTION**

Determine the current in the capacitor branch by the superposition theorem.



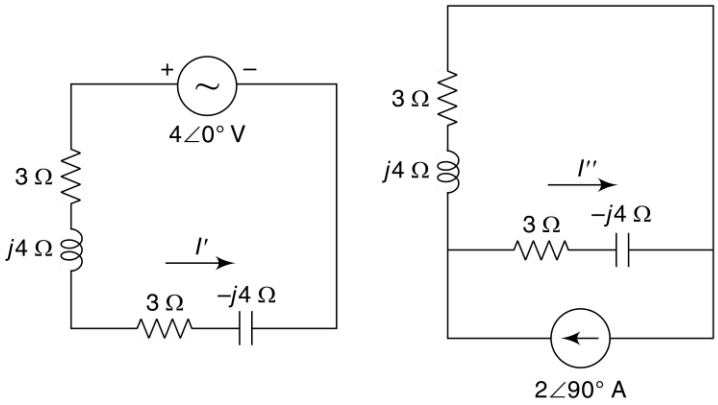

$Z_1 = 3 + j4\ \Omega$ (left branch); $Z_2 = 3 - j4\ \Omega$ (capacitor branch); current source $2\angle 90°$ A.

**TRY IT FIRST**

Which way does each source see $Z_1$ and $Z_2$: in series, or in parallel?

**HINT: WHAT'S NEW**

- **Source with an angle:** $2\angle 90°$ A = j2 A. Convert to rectangular form before multiplying.
- **Polar form with a negative real part:** a calculator's $\tan^{-1}$ lands in the wrong quadrant; add 180°.




(i) $4\angle 0°$ V alone · (ii) $2\angle 90°$ A alone

**Given:** $4\angle 0°$ V, $2\angle 90°$ A, $Z_1 = 3 + j4\ \Omega$, $Z_2 = 3 - j4\ \Omega$.

**To find:** I in the capacitor branch (left → right).

**Case (i): $4\angle 0°$ V alone** (current source opened). $Z_1$, $Z_2$ in series:

$$I' = \frac{4\angle 0°}{(3 + j4) + (3 - j4)} = \frac{2}{3}\angle 0°\ \text{A}$$

**Case (ii): 2∠90° A alone** (voltage source shorted). $Z_1 \parallel Z_2$ across the source:

$$I'' = j2 \times \frac{3 + j4}{6} = -\frac{4}{3} + j1 \text{ A}$$

**Superposition:** $I = \frac{2}{3} - \frac{4}{3} + j1 = -\frac{2}{3} + j1$ A

$$|I| = \sqrt{((2/3)^2 + 1)} = 1.20 \text{ A}, \qquad \theta = 180° - \tan^{-1}(1.5) = 123.7°$$

**ANSWER**

**I = 1.2∠123.7° A** (= book)

**SIDE NOTE**

$Z_1 + Z_2 = 6\ \Omega$ is purely real (+j4 and −j4 cancel), which is why both cases are clean. Equal and opposite reactances in series act like a short: **series resonance**, met later.

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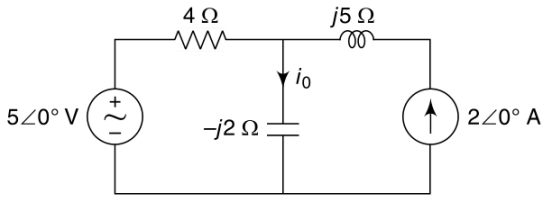
## A5 · Problem 5.5

Superposition · AC, DC + AC **teach**

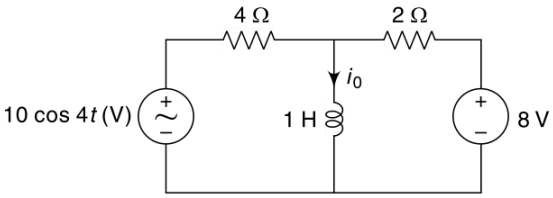
**QUESTION**

Find the current $i_0$ using the superposition theorem, in each of circuits (a), (b) and (c).

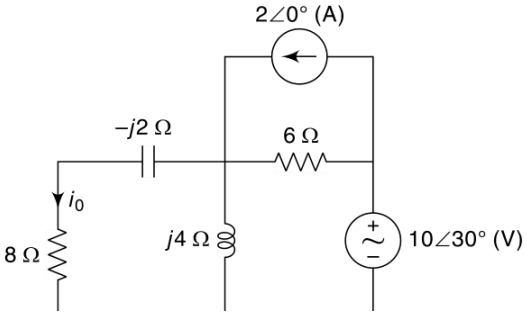
(a)




(b)




(c)



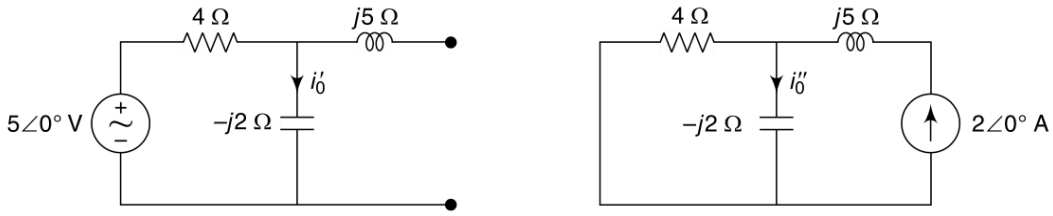
**TRY IT FIRST**

(a) is quick practice. In (b), stop and ask: what does an inductor look like to the 8 V **DC** source? In (c), redraw the circuit after switching off each source before you calculate anything.

**HINT: WHAT'S NEW**

- **(b) Two different frequencies (DC and 10 cos 4t).** Each case is solved **at its own frequency**: at DC ($\omega = 0$), $X_L = \omega L = 0$, so the **inductor is a short**; at $\omega = 4$, it is j4 Ω. The two answers are added **in the time domain**, never as phasors, because phasors only add when they share one frequency.
- **(c) Redraw after switching off.** Once the voltage source is shorted, the 6 Ω and the current source both run to ground and everything becomes parallel.

**(a) Given:** 5∠0° V, 2∠0° A, 4 Ω, −j2 Ω, j5 Ω. **To find:** $i_0$ (down the capacitor).




(i) voltage source alone · (ii) current source alone

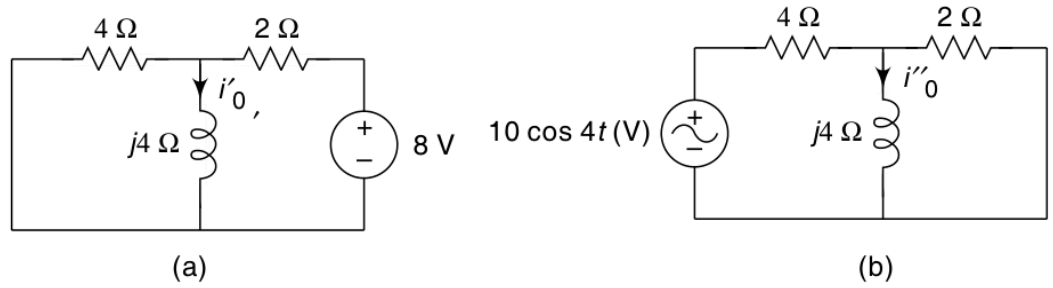
$$5\angle 0° \text{ alone:} \qquad i_0' = \frac{5}{4 - j2} = 1 + j0.5 \text{ A}$$

$$2\angle 0° \text{ alone (j5 is in series with the source, so irrelevant):} \qquad i_0'' = 2 \times \frac{4}{4 - j2} = 1.6 + j0.8 \text{ A}$$

**ANSWER**

**(a) $i_0 = 2.6 + j1.3 = 2.91\angle 26.57°$ A** (= book, 2.9∠26.56° A)

**(b) Given:** 10 cos 4t V, 8 V DC, 4 Ω, 1 H, 2 Ω. **To find:** $i_0(t)$ (down the inductor).




The book's sub-circuits (a) DC alone, (b) AC alone. In (a) the inductor must be read as a short circuit.

**8 V DC alone** (AC source shorted; $\omega = 0$, so the inductor is a short):

$$i_0' = \frac{8}{2} = 4\ \text{A} \qquad \text{(the 4 Ω has 0 V across it)}$$

**10 cos 4t alone** (8 V shorted; ω = 4 rad/s, $Z_L$ = j4 Ω):

$$Z = 4 + \frac{j4 \times 2}{2 + j4} = 5.6 + j0.8\ \Omega, \qquad I = \frac{10\angle 0°}{Z} = 1.75 - j0.25\ \text{A}$$

$$i_0'' = I \times \frac{2}{2 + j4} = 0.25 - j0.75 = 0.791\angle -71.57°\ \text{A}$$

**Superposition (time domain):**

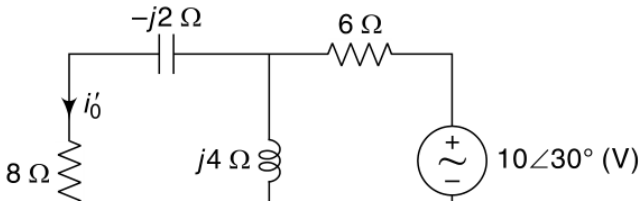
**ANSWER**

**(b) $i_0(t) = 4 + 0.791 \cos(4t - 71.57°)$ A**

**BOOK ERROR IN 5.5(B)**

The book treats the inductor as j4 Ω for the **DC** source too, and then adds the DC and AC results as if both were phasors, printing 2.63∠−63.58° A. Both steps are wrong: at DC an inductor is a short, and quantities at different frequencies cannot be added as phasors. The answer above was confirmed by a node solve of each case.

**(c) Given:** 10∠30° V, 2∠0° A, 8 Ω, −j2 Ω, j4 Ω, 6 Ω. **To find:** $i_0$ (down the 8 Ω).

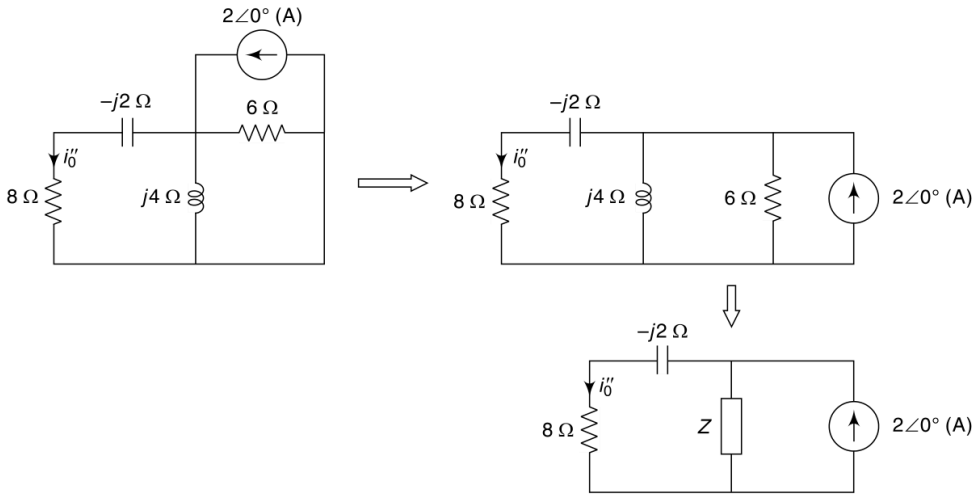



(i) voltage source alone

**10∠30° V alone** (current source opened):

$$Z = 6 + j4 \parallel (8 - j2) = 6 + (1.882 + j3.529) = 7.882 + j3.529\ \Omega, \qquad I = \frac{10\angle 30°}{Z} = 1.158\angle 5.88°\ \text{A}$$

$$i_0' = I \times \frac{j4}{(8 - j2) + j4} = I \times \frac{j4}{8 + j2} = 0.562\angle 81.84°\ \text{A}$$




(ii) current source alone, redrawn: the 6 Ω and the source now both go to ground

**2∠0° A alone** (voltage source shorted). Z = j4 ∥ 6 = 1.846 + j2.769 Ω:

$$i_0'' = 2 \times \frac{Z}{(8 - j2) + Z} = \frac{j24}{28 + j22} = 0.674\angle 51.84^\circ \text{ A}$$

$$i_0 = (0.080 + j0.556) + (0.416 + j0.530) = 0.496 + j1.086 \text{ A}$$

**ANSWER**

**(c) $i_0 = 1.19\angle 65.45^\circ$ A**

**BOOK ERROR IN 5.5(C)**

The book prints 0.81∠39.38° A. Two slips: in case (i) it divides the current with (8 − j2)/(8 + j2) instead of j4/(8 + j2) (the current into a branch is set by the **other** branch's impedance); in case (ii) it expands (8 − j2)(3 + j2) + j12 as 12 + j11 instead of 28 + j22. The answer above was confirmed by a full node solve.

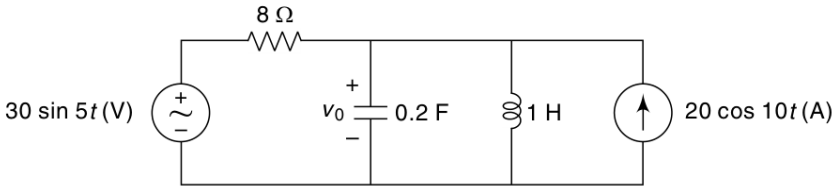
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## A6 · Problem 5.6

Superposition · two frequencies **new idea**

**QUESTION**

Find $v_0$ using the superposition theorem.

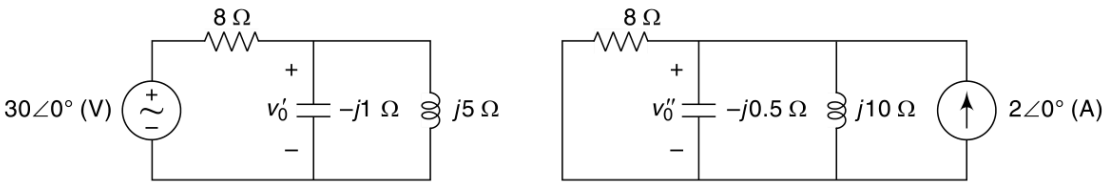



**TRY IT FIRST**

Work out $X_C$ and $X_L$ separately for ω = 5 and ω = 10 before drawing anything.

**HINT: WHAT'S NEW**

- **Reactances change with frequency.** $X_C = \frac{1}{\omega C}$ and $X_L = \omega L$ must be recomputed for each source's own ω.
- **Mixed sin and cos:** keep each term in the function it started with (sin for the voltage source, cos for the current source).




(i) voltage source alone, ω = 5 · (ii) current source alone, ω = 10

**Given:** 30 sin 5t V with 8 Ω; 0.2 F, 1 H, 20 cos 10t A. **To find:** $v_0(t)$.

**30 sin 5t alone** (ω = 5; current source opened): $X_C = \frac{1}{5 \times 0.2} = 1\ \Omega \to -j1\ \Omega$; $X_L = 5 \times 1 = 5\ \Omega \to j5\ \Omega$.

$$\frac{30 - v'}{8} = \frac{v'}{-j1} + \frac{v'}{j5} \quad \Rightarrow \quad v' = \frac{30}{8(0.125 + j0.8)} = 4.631\angle -81.12°\ \text{V}$$

**20 cos 10t alone** (ω = 10; voltage source shorted): $X_C = \frac{1}{10 \times 0.2} = 0.5\ \Omega \to -j0.5\ \Omega$; $X_L = 10\ \Omega \to j10\ \Omega$.

$$20 = v''\left(\frac{1}{8} + \frac{1}{-j0.5} + \frac{1}{j10}\right) = v''(0.125 + j1.9) \quad \Rightarrow \quad v'' = 10.50\angle -86.24°\ \text{V}$$

**ANSWER**

**$v_0(t) = 4.631 \sin(5t - 81.12°) + 10.50 \cos(10t - 86.24°)$ V**

**BOOK INCONSISTENCY IN 5.6**

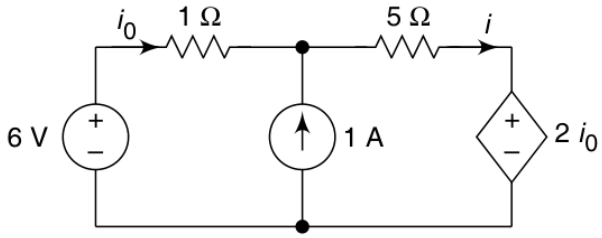
The figure gives the current source as **20** cos 10t A, but the book's solution uses **2**∠0° A and prints 1.051 cos(10t − 86.24°) for the second term. The answer above follows the figure. If your source is 2 A, the second term is 1.051 cos(10t − 86.24°) V; the first term is unchanged.

## A7 · Problem 5.7

Superposition · current-controlled source **new idea**

**QUESTION**

Find $i_0$ and i from the circuit of the figure using the superposition theorem.



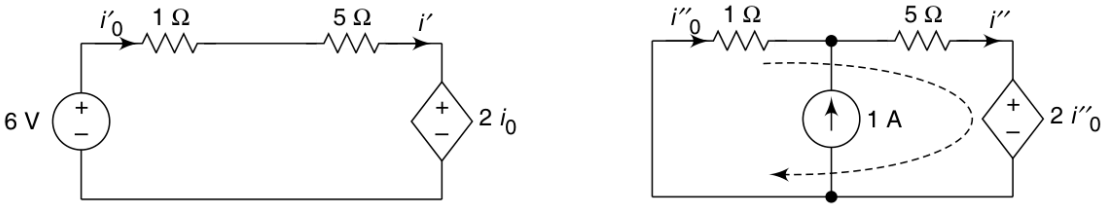

$2i_0$ is a current-controlled voltage source.

**TRY IT FIRST**

In each case, the dependent source's value is 2 × (that case's $i_0$). Write KCL at the middle node and one KVL around the outer loop.

**HINT: WHAT'S NEW**

- **Controlled by a current, not a voltage** (5.3 was voltage-controlled). The rule is the same: keep the source, and re-find its controlling current $i_0$ in each case.




(i) 6 V alone · (ii) 1 A alone

**Given:** 6 V, 1 A, 1 Ω, 5 Ω, CCVS $2i_0$. **To find:** $i_0$, i.

**6 V alone** (1 A opened): one loop, so $i_0' = i'$.

$$6 = 1\cdot i' + 5\cdot i' + 2i_0' = 8i' \quad \Rightarrow \quad i' = i_0' = 0.75 \text{ A}$$

**1 A alone** (6 V shorted):

$$\text{KCL:} \quad i'' = 1 + i_0'' \qquad \text{outer KVL:} \quad 1\cdot i_0'' + 5i'' + 2i_0'' = 0$$

$$\Rightarrow \quad 3i_0'' + 5(1 + i_0'') = 0$$

$$8i_0'' = -5 \quad \Rightarrow \quad i_0'' = -0.625 \text{ A}, \quad i'' = 0.375 \text{ A}$$

**ANSWER**

**$i = 0.75 + 0.375 = 1.125$ A, $i_0 = 0.75 - 0.625 = 0.125$ A**

**BOOK ERROR IN 5.7**

The book sets up the same equation, then solves $8i_0'' = -5$ as $i_0'' = -5/4 = -1.25$ A, and prints $i = 0.5$ A, $i_0 = -0.5$ A. The correct value is $-5/8 = -0.625$ A. Direct check: KCL $i = i_0 + 1$ and KVL $6 - i_0 - 5i = 2i_0$ give $i_0 = 1/8$ A ✓.

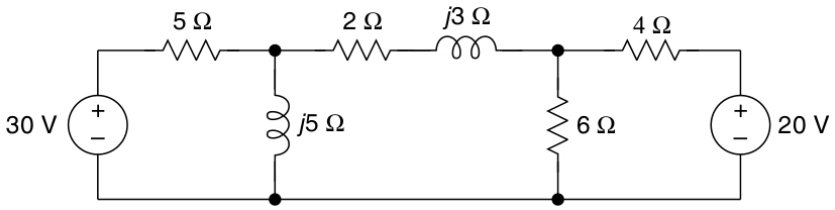
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## A8 · Problem 5.8

Superposition · AC **practice**

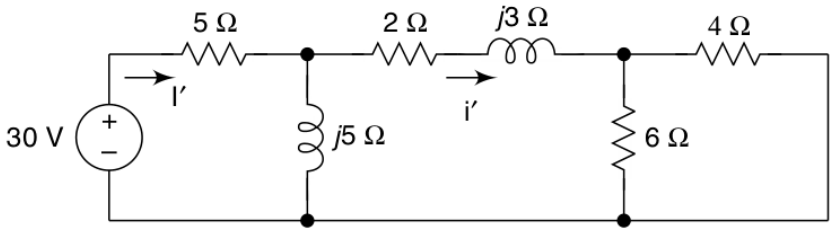
**QUESTION**

Using the superposition theorem, calculate the current through the (2 + j3) Ω impedance branch.




**TRY IT FIRST**

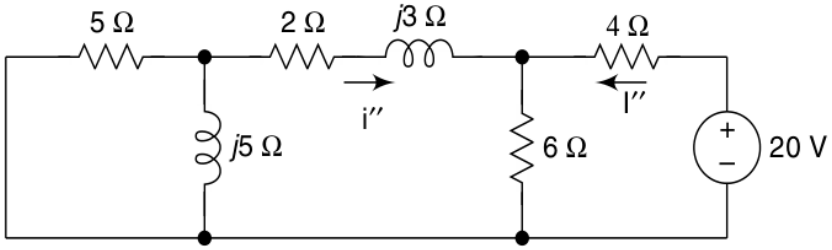
Series-parallel reduction, twice. Watch which way the branch current flows in case II.




Case I: 30 V alone

**Case I: 30 V alone** (20 V shorted). Branch: $Z_b = (2 + j3) + (6 \parallel 4) = 4.4 + j3$ Ω.

$$Z = 5 + \frac{Z_b \times j5}{Z_b + j5} = 6.32 + j2.60\ \Omega, \qquad I' = \frac{30}{Z} = 4.06 - j1.67\ \text{A}, \qquad i' = I' \times \frac{j5}{Z_b + j5} = 2.389 + j0.270\ \text{A}$$




Case II: 20 V alone

**Case II: 20 V alone** (30 V shorted). Branch: $Z_a = (5 \parallel j5) + 2 + j3 = 4.5 + j5.5$ Ω.

$$Z = 4 + \frac{6Z_a}{Z_a + 6} = 7.31 + j1.41\ \Omega, \qquad I'' = \frac{20}{Z} = 2.638 - j0.509\ \text{A}$$

$$i'' = -I'' \times \frac{6}{Z_a + 6} = -1.063 + j0.848\ \text{A} \qquad \text{(it flows right} \rightarrow \text{left, against i)}$$

**ANSWER**

**i = i′ + i″ = 1.326 + j1.117 = 1.73∠40.13° A** (= book)

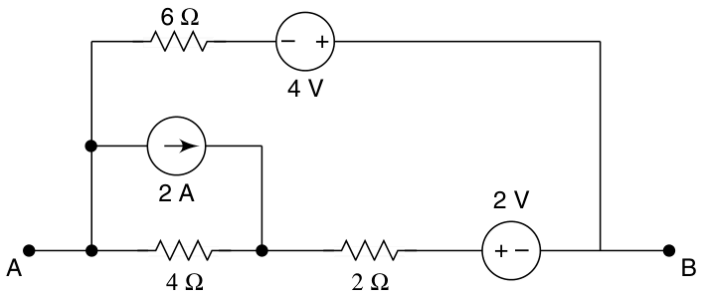
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## A9 · Problem 5.9

Superposition · three sources, open terminals **new idea**

**QUESTION**

Using the superposition theorem, find $V_{AB}$.

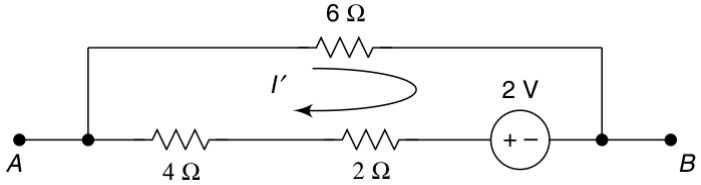



**TRY IT FIRST**

Three sources, so three cases. Nothing is connected across A–B, so no current leaves at A or B.

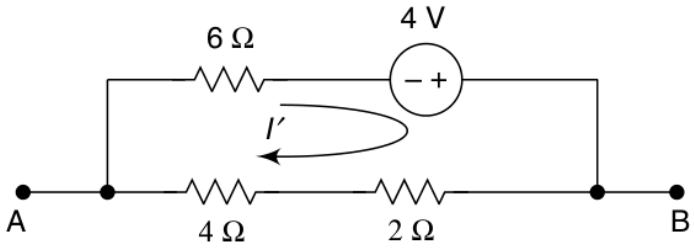
**HINT: WHAT'S NEW**

- **Three sources:** same method, one more case.
- **Open terminals:** $V_{AB}$ is the voltage across a gap. With no load, the only path is the loop 4 Ω + 2 Ω + 6 Ω, and $V_{AB}$ is the drop across the 6 Ω.
- **Source transformation inside a case:** 2 A in parallel with 4 Ω equals an 8 V source in series with 4 Ω, which makes case III a single loop.



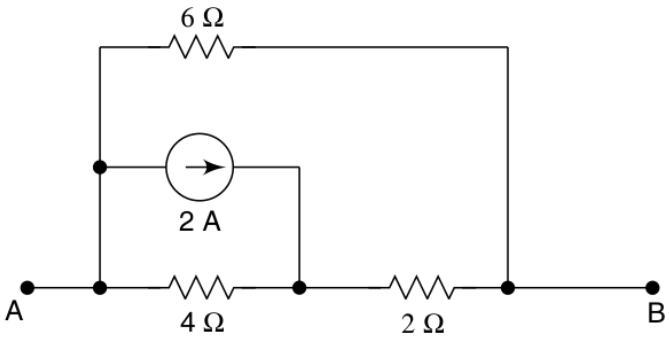

Case I: 2 V alone

Case I (2 V): $I' = \dfrac{2}{4+2+6} = \dfrac{1}{6}$ A, $V_{AB}' = 6I' = 1$ V

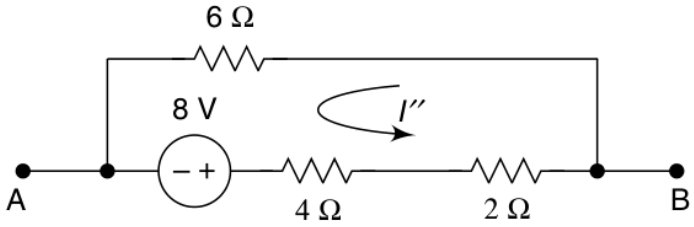



Case II: 4 V alone

Case II (4 V): $I'' = \dfrac{4}{12} = \dfrac{1}{3}$ A, $V_{AB}'' = -6I'' = -2$ V




Case III: 2 A alone




Case III after source transformation: 2 A ∥ 4 Ω → 8 V in series with 4 Ω

Case III (2 A): $I''' = \dfrac{8}{12} = \dfrac{2}{3}$ A, $V_{AB}''' = -6I''' = -4$ V

(In case I the current runs through the 6 Ω from A's side to B's side, so A is higher; in cases II and III it runs the other way, hence the minus signs.)

**ANSWER**

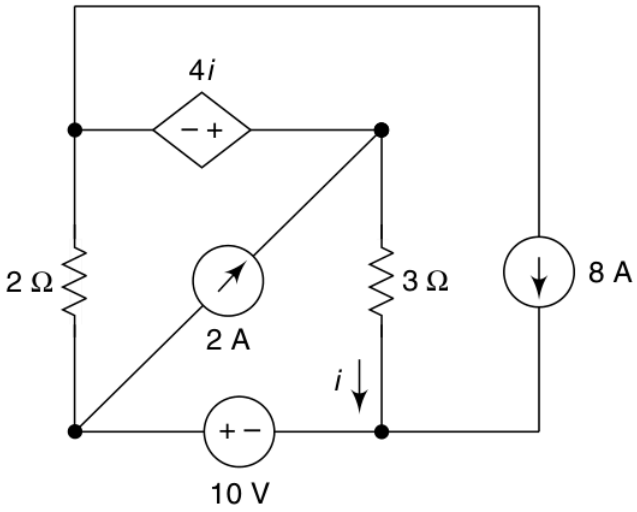
**$V_{AB} = 1 - 2 - 4 = -5$ V** (= book)

## A10 · Problem 5.10

Superposition · dependent source, three sources **new idea**

**QUESTION**

Find the current i in the circuit shown using the superposition theorem.



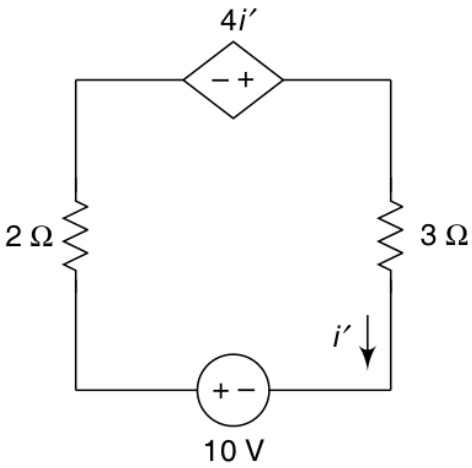

4i is a current-controlled voltage source; i flows down the 3 Ω.

**TRY IT FIRST**

Case I is one loop. In case II, look for a resistor in series with the dependent source.

**HINT: WHAT'S NEW**

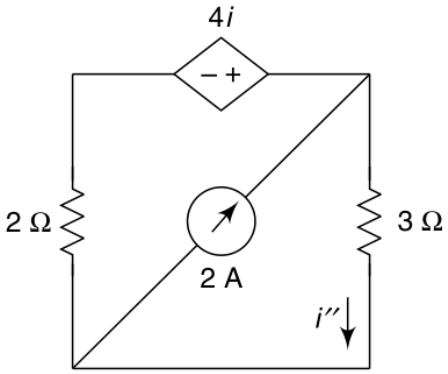
- **Source transformation works on dependent sources too.** In case II, the 4i source in series with the 2 Ω becomes a dependent current source 4i/2 = **2i** in parallel with 2 Ω.
- **A current source inside a mesh fixes that mesh current** (case III: the 8 A source sets the mesh difference I − 8).



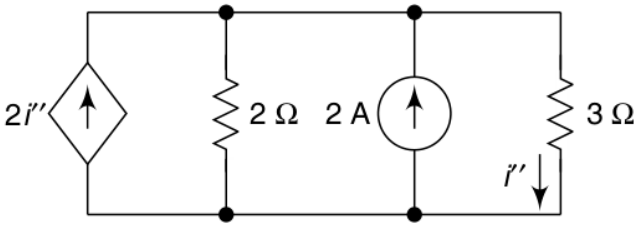

Case I: 10 V alone

**Case I: 10 V alone** (2 A and 8 A opened). Loop current i′ clockwise:

$$-4i' + 3i' - 10 + 2i' = 0 \qquad \Rightarrow \qquad i' = 10\ \text{A}$$



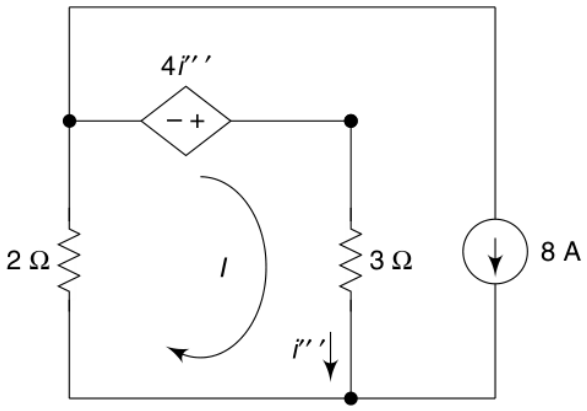

Case II: 2 A alone




Case II after converting 4i″ + 2 Ω into 2i″ ‖ 2 Ω

**Case II: 2 A alone** (10 V shorted, 8 A opened). Total (2 + 2i″) splits between 2 Ω and 3 Ω:

$$i'' = \frac{2}{2+3}(2 + 2i'') \qquad \Rightarrow \qquad 5i'' = 4 + 4i'' \qquad \Rightarrow \qquad i'' = 4\ \text{A}$$




Case III: 8 A alone

**Case III: 8 A alone** (10 V shorted, 2 A opened). Mesh current I; the 3 Ω carries i‴ = I − 8:

$$-4i''' + 3(I - 8) + 2I = 0, \qquad I = i''' + 8 \qquad \Rightarrow \qquad -4i''' + 3i''' + 2i''' + 16 = 0$$
$$\Rightarrow \qquad i''' = -16\ \text{A}$$

**ANSWER**

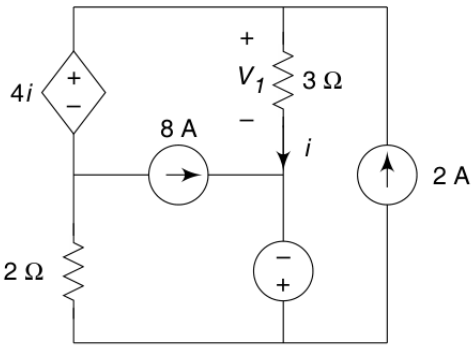
**i = 10 + 4 − 16 = −2 A** (= book)

## A11 · Problem 5.11

Superposition · dependent source, three sources **practice**

**QUESTION**

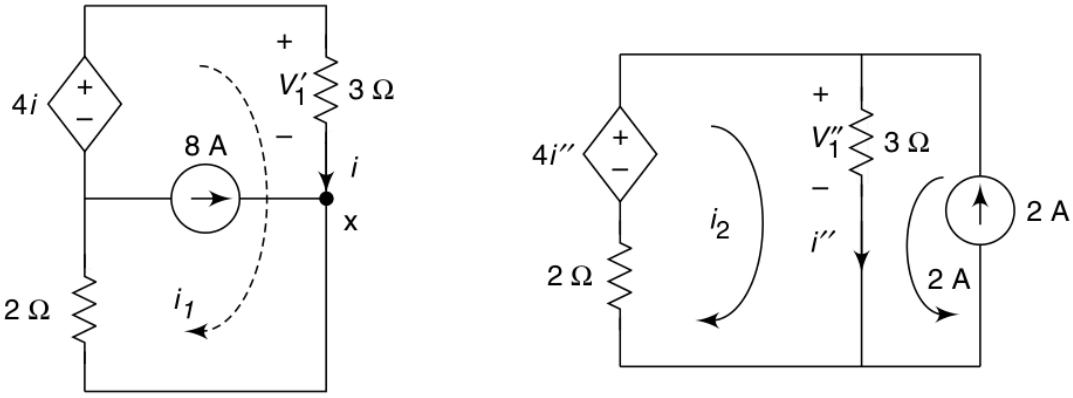
Using the superposition theorem, determine $V_1$, the voltage across the 3 Ω resistor.




The figure prints no value on the bottom-middle voltage source; the book's solution uses 10 V.

**TRY IT FIRST**

Three cases again; the dependent source 4i stays in all three. Case I has a supermesh around the 8 A source.



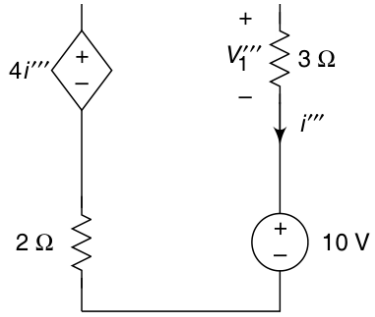

Case I: 8 A alone · Case II: 2 A alone

**Case I: 8 A alone.** Supermesh KVL and KCL at node x:

$$3i' + 2i_1 - 4i' = 0 \Rightarrow i_1 = \frac{i'}{2}; \qquad i_1 = 8 + i' \Rightarrow \frac{i'}{2} = 8 + i' \Rightarrow i' = -16 \text{ A}, \qquad V_1' = 3i' = -48 \text{ V}$$

**Case II: 2 A alone.** Mesh current $i_2$ on the left; the 3 Ω carries $i'' = i_2 + 2$:

$$3(i_2 + 2) + 2i_2 - 4i'' = 0 \Rightarrow 5i_2 + 6 - 4(i_2 + 2) = 0 \Rightarrow i_2 = 2 \text{ A}, \qquad i'' = 4 \text{ A}, \qquad V_1'' = 12 \text{ V}$$




Case III: 10 V alone

**Case III: 10 V alone.** One loop:

$$3i''' - 10 + 2i''' - 4i''' = 0 \Rightarrow i''' = 10\ \text{A}, \qquad V_1''' = 30\ \text{V}$$

**ANSWER**

**$V_1 = -48 + 12 + 30 = -6$ V** (= book)

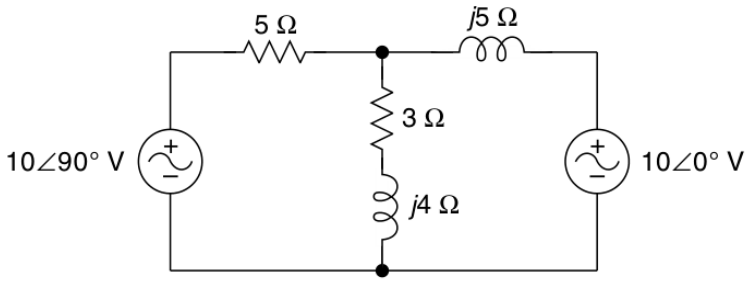
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## A12 · Problem 5.12

Superposition · AC **new idea**

**QUESTION**

Calculate the current through the impedance (3 + j4) Ω using the superposition theorem.

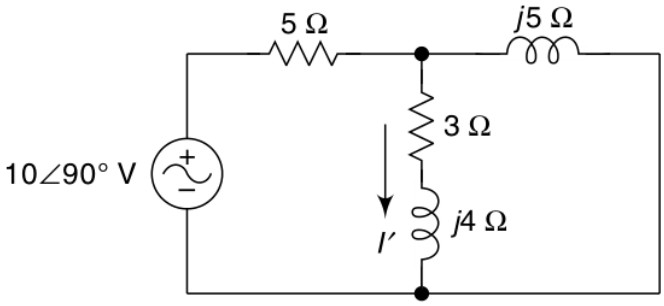



**TRY IT FIRST**

Two voltage sources, same frequency. Get each case's current as a fraction before simplifying.

**HINT: WHAT'S NEW**

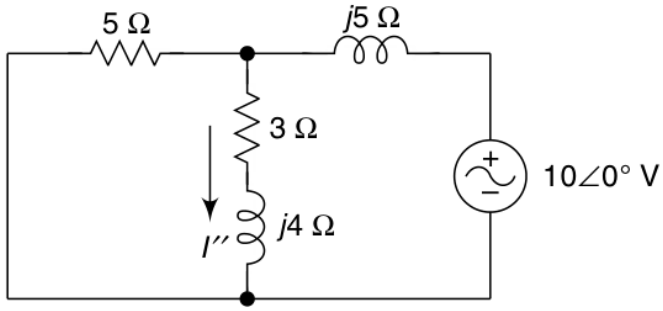
- **The two contributions can cancel.** Here they come out equal and opposite, so the branch current is zero. A zero answer is not an error; check it with one node equation.




10∠90° V alone

10∠90° alone: $I = \dfrac{j10}{5 + \dfrac{(3 + j4)j5}{3 + j9}} = \dfrac{j10(3 + j9)}{-5 + j60}$, $I' = I \times \dfrac{j5}{3 + j9} = \dfrac{-10}{-1 + j12} =$

$0.069 + j0.828$ A




10∠0° V alone

10∠0° alone: $I = \dfrac{10}{j5 + \dfrac{5(3 + j4)}{8 + j4}} = \dfrac{10(8 + j4)}{-5 + j60}$, $I'' = I \times \dfrac{5}{8 + j4} = \dfrac{10}{-1 + j12} =$

$-0.069 - j0.828$ A

**ANSWER**

**I = I′ + I″ = 0 A** (= book)

**CHECK**

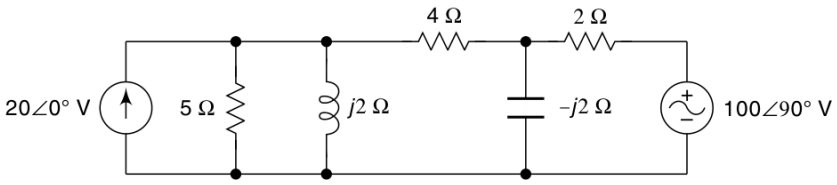
Node equation at the top node: $V(\frac{1}{5} + \frac{1}{3 + j4} + \frac{1}{j5}) = \frac{j10}{5} + \frac{10}{j5} = j2 - j2 = 0$, so V = 0 and no current flows in 3 + j4 ✓.

## A13 · Problem 5.13

Superposition · AC **practice**

**QUESTION**

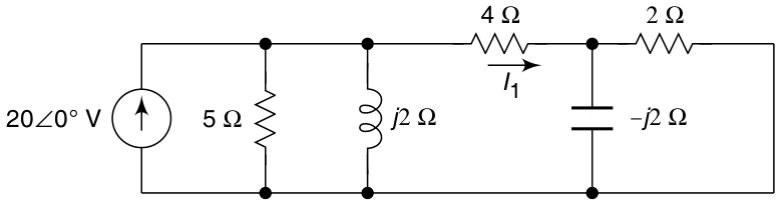
Using the superposition theorem, determine the current in the 4 Ω resistor.




The book labels the current source "20∠0° V"; it is a current source (arrow), 20∠0° A.

**TRY IT FIRST**

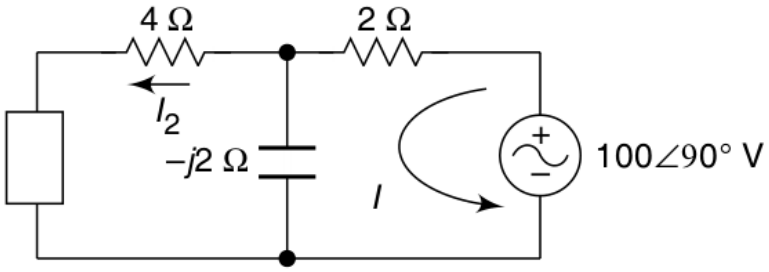
Combine the parallel pairs first: 5 Ω ∥ j2 Ω and 2 Ω ∥ −j2 Ω.




Case I: 20∠0° A alone

**Case I: 20∠0° A alone** (voltage source shorted). $Z_1 = 5 \parallel j2 = 0.690 + j1.724\ \Omega$; $Z_2 = 2 \parallel (-j2) = 1 - j1\ \Omega$.

$$I_1 = 20 \times \frac{Z_1}{Z_1 + 4 + Z_2} = 3.145 + j5.660\ \text{A} \qquad \text{(left} \rightarrow \text{right)}$$




Case II: 100∠90° V alone

**Case II: 100∠90° V alone** (current source opened). $(Z_1 + 4) \parallel (-j2) = 0.850 - j1.950\ \Omega$:

$$I_s = \frac{j100}{2 + (0.850 - j1.950)} = -16.35 + j23.90\ \text{A}, \qquad V_B = I_s(0.850 - j1.950) = 32.70 + j52.20\ \text{V}$$

$$I_2 = \frac{V_B}{Z_1 + 4} = 9.748 + j7.547\ \text{A} \qquad \text{(right} \rightarrow \text{left)}$$

**ANSWER**

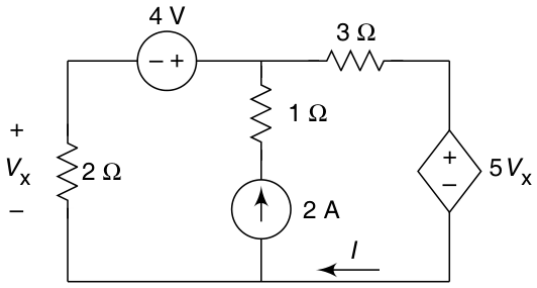
**$I = I_1 - I_2 = -6.604 - j1.887 = 6.87\angle -164.05°$ A (left → right)**, i.e. 6.87 A flowing right → left (= book, 6.89∠−163.67° A, to rounding)

## A14 · Problem 5.14

Superposition · dependent source **practice**

**QUESTION**

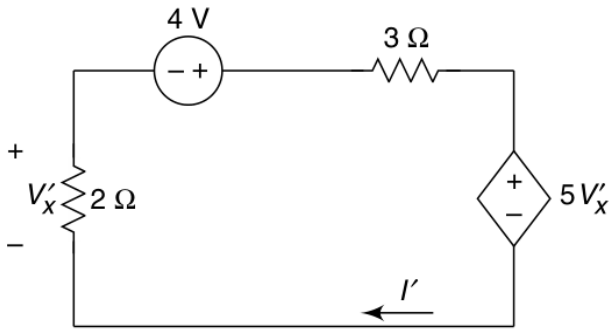
Find I in the figure using the superposition theorem.




$5V_x$ is a voltage-controlled voltage source; $V_x$ is across the 2 Ω (+ at top).

**TRY IT FIRST**

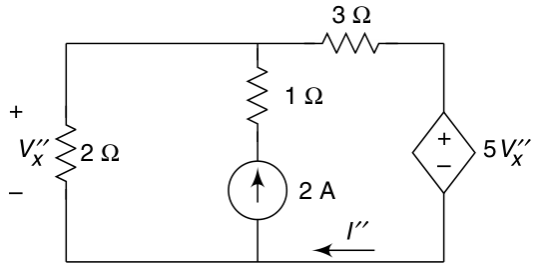
Same idea as 5.3. In case II, the 4 V short joins the two top-left nodes; write one KCL there.




Case I: 4 V alone

**Case I: 4 V alone** (2 A opened). One loop; I′ flows up the 2 Ω, so $V_x' = -2I'$:

$$-4 + 3I' + 5V_x' - V_x' = 0 \Rightarrow 3I' + 4(-2I') = 4 \Rightarrow I' = -0.8 \text{ A}$$




Case II: 2 A alone

**Case II: 2 A alone** (4 V shorted). KCL at the top node (voltage $V_x''$):

$$2 = \frac{V_x''}{2} + \frac{V_x'' - 5V_x''}{3} \Rightarrow V_x'' = -2.4 \text{ V}, \qquad I'' = \frac{V_x'' - 5V_x''}{3} = 3.2 \text{ A}$$

**ANSWER**

**I = −0.8 + 3.2 = 2.4 A** (= book)